

SOURAV YADAV

Software Development Engineer | Full-Stack Developer | Backend Engineer

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📍 Madhya Pradesh, India

🔗 [GitHub](#) | [Portfolio](#) | [LeetCode](#) | [LinkedIn](#) | [X\(Twitter\)](#)



PROFESSIONAL SUMMARY

Software engineer with strong foundations in Data Structures & Algorithms, backend engineering, distributed systems, and full-stack application development.

Experienced designing and building scalable software systems, backend services, transaction processing platforms, routing infrastructure, and developer tooling using C++, TypeScript, React, Node.js, Express.js, SQL, and Rust. Passionate about solving complex engineering problems through clean architecture, testing, debugging, performance optimization, and maintainable software design.

TECHNICAL SKILLS

- **Languages:** C++, JavaScript, TypeScript, Rust, Python (Basic), Solidity.
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design.
- **Backend:** Node.js, Express.js, REST APIs, PostgreSQL, MySQL, Redis, Prisma ORM, JWT Authentication.
- **Frontend:** React.js, TypeScript, JavaScript (ESM), HTML5, CSS3, Tailwind CSS.
- **Tools & Platforms:** Git, GitHub, Linux, Postman, Docker, Vercel, Netlify.
- **Additional Technologies:** Ethers.js, Viem, Wagmi, Web3.js, The Graph, Chainlink (CRE, Price Feed, Automation, Functions, VRF), Foundry, Hardhat.

PROJECTS

PreFlight — Pre-Transaction Risk Analysis Platform

[GitHub Repository](#)

Solidity · Foundry · Chainlink CRE · React · Viem · Browser Extension

- Designed and developed a **modular transaction analysis platform** that validates user transactions before execution through real-time simulation and automated security analysis.
- Architected backend workflows for transaction processing, state validation, and risk evaluation while integrating frontend applications with distributed smart contract services.
- Built **reusable React components, TypeScript modules, and contract interaction layers** following modular architecture and maintainable software engineering practices.

Protocol Invariant Checker — Rust CLI Framework

[GitHub Repository](#)

Rust · CLI · Security Tooling · Protocol Analysis

- Developed a **modular command-line application in Rust** for automated protocol analysis and structured report generation.
- Designed extensible software architecture using **trait-based abstractions, enabling reusable protocol implementations.**
- Implemented validation pipelines, reporting modules, and comprehensive unit tests following clean software engineering principles.

SettleOne — Payment Commitment Platform

[GitHub Repository](#)

Solidity · Foundry · Chainlink · React · NodeJs · ExpressJs · Viem

- Designed and developed a **full-stack payment commitment platform supporting secure buyer-seller transactions through escrow-backed workflows.**

- Architected backend APIs for payment lifecycle management, transaction verification, dispute resolution, and settlement.
- Built scalable frontend interfaces and modular backend services with emphasis on reliability, maintainability, and secure transaction processing.

DexGateway — Multi-Network Routing Infrastructure

[GitHub Repository](#)

[Solidity](#) · [Foundry](#) · [React](#) · [NodeJs](#) · [ExpressJs](#) · [TypeScript](#) · [The Graph](#)

- Built a **routing infrastructure integrating multiple exchanges across various blockchain networks** through a unified execution engine.
- Implemented route optimization algorithms based on liquidity discovery, pricing, and execution cost.
- Developed backend services, transaction execution pipelines, and frontend workflows using modular software architecture.

SOFTWARE ENGINEERING EXPERIENCE

Independent Software Security Researcher

github.com/Sourav-III/TBPL/audits

[Sherlock](#) | [Code4rena](#) | [Cantina](#)

July, 2025 – present

- Reviewed and analyzed **20+ complex distributed software systems** to identify logical flaws, edge cases, and reliability issues through systematic code analysis.
- Developed **reproducible proof-of-concept implementations and testing environments** to validate software behavior under adversarial scenarios.
- Strengthened expertise in **debugging, software architecture analysis, testing methodologies, and large-scale codebase comprehension** through extensive code reviews.

EDUCATION

Indian Institute of Information Technology (IIIT), Bhopal

Class of 2027

B.Tech — Electronics & Communication Engineering

CGPA: 7.57 / 10 | Expected Graduation: 2027

Maharishi Vidya Mandir, Mandla

Higher Secondary Education (CBSE, Class XII) — **89.4%**

ACHIEVEMENTS

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- **LeetCode Knight (1868 Rating)** — Solved **500+ DSA problems** with strong algorithmic problem-solving skills.
 - **HackVision 2026** — **Top 100 finalist team** among **400+ teams** in a national virtual hackathon.
 - **Smart India Hackathon 2024** — National-level participant and team lead.
 - **AlgoUniversity Tech Fellowship** — Selected for **2024 & 2025 cohorts**.
 - **HER DAO Rust Cohort 1 Graduate**.
 - **Cyfrin Updraft Certified** — Advanced Solidity development, smart contract security, protocol testing, and DeFi protocol engineering ([Certification Profile](#)).